

M3Jar — Phase 4D Final Report

Jar Goals + Financial Transparency

Delivered: August 5, 2026

HEAD === origin/main | All 501 tests passing | 3 typechecks clean

1. Executive Summary

Phase 4D adds jar goal planning and full financial transparency to the M3Jar savings platform. Organizers can define, order, and archive named funding goals that are automatically waterfall-allocated from real ledger principal. A new financial-summary endpoint surfaces privacy-scoped breakdowns: organizers see exact figures per member; regular members see their own exact data plus only a status label and percent-funded for others. All 501 backend tests pass on a single test run. Typechecks are clean across api-server, mobile, and lib/db. The commit is live on origin/main.

2. Architecture Decisions (Pre-Approved)

2a. Ledger-backed totalSavedCents

The old getTotalSaved() queried the contributions table. Phase 4D replaces it with getJarSavedPrincipalCents() that reads ledger_entries. The formula:

```
savedPrincipalCents = refundablePrincipalCents + committedPrincipalCents
```

Rows with type IN ('refund_pending', 'refunded', 'drip_jar_fee', 'processing_fee') and simulated contribution rows are excluded. The same formula is used everywhere: the progress bar, the jar health endpoint, the financial summary, and the goal waterfall.

2b. sort_order Strategy

Sequential integers (1, 2, 3...). On every reorder, all active goals for the jar are fully reassigned inside a single transaction. No fractional / float ordering. No DB UNIQUE constraint on sort_order (duplicates during mid-transaction state are harmless and the final state is always correct).

2c. Goal Deletion Policy

Hard delete when jar.status IN ('Draft', 'Inviting') — goal never launched, no history needed. Soft archive (status='archived', archivedAt set) once the jar has launched. Archived goals are excluded from the waterfall calculation and from the active target sum used in constraint checks.

3. Schema & Migration

Migration 0018 — jar_goals

lib/db/drizzle/0018_phase4d_jar_goals.sql

New table: jar_goals

id	UUID PRIMARY KEY DEFAULT gen_random_uuid()
jar_id	UUID NOT NULL REFERENCES jars(id) ON DELETE CASCADE
name	TEXT NOT NULL
description	TEXT
target_principal_cents	INTEGER NOT NULL CHECK (target_principal_cents > 0)
sort_order	INTEGER NOT NULL DEFAULT 0
status	TEXT NOT NULL DEFAULT 'active' CHECK (status IN ('active', 'archived'))
created_at	TIMESTAMPTZ NOT NULL DEFAULT now()
updated_at	TIMESTAMPTZ NOT NULL DEFAULT now()
archived_at	TIMESTAMPTZ

Index: jar_goals_jar_id_status_sort_idx ON (jar_id, status, sort_order)

The journal entry (idx:18) was registered in meta/_journal.json. Migration applied to the live database.

lib/db was rebuilt (pnpm tsc -p lib/db/tsconfig.json) and all type exports verified.

4. API Surface

Goals Router — artifacts/api-server/src/routes/goals.ts

GET /api/jars/:jarId/goals active member or organizer — waterfall-allocated goals list
GET /api/jars/:jarId/goals?includeArchived=true same, plus archived goals in a separate archivedGoals array
POST /api/jars/:jarId/goals organizer only — create goal; sum constraint enforced FOR UPDATE
PATCH /api/jars/:jarId/goals/:goalId organizer only — update name/desc/target; sum constraint re-checked
POST /api/jars/:jarId/goals/:goalId/archive organizer only — hard delete (Draft/Inviting) or soft archive (Saving+)
POST /api/jars/:jarId/goals/reorder organizer only — full reassignment of sort_order 1...N; rejects incomplete/invalid lists

Financial Summary Router — artifacts/api-server/src/routes/financial-summary.ts

GET /api/jars/:jarId/financial-summary

Requires active member or organizer. Returns jar-level principal breakdown, goal waterfall, and per-member figures. Privacy model:

- Organizer view: exact contributedPrincipalCents, refundable, committed, refundPending, refunded, fees for every member
- Member view: own exact figures (isOwnData=true) + statusLabel + savedPercent only for all others
- Invariant check: committedPrincipalCents "d savedPrincipalCents always verified and surfaced as invariantHolds

jars.ts — breaking changes

getTotalSaved() removed. All three call-sites (buildJarSummary, GET /jars/:jarId, GET /jars/:jarId/health) replaced with getJarSavedPrincipalCents(jarId) from financial-balance.ts. PATCH /jars/:jarId now validates active goal sum "d new goalAmountCents when the organizer reduces the jar target.

5. Pure Waterfall Function

artifacts/api-server/src/lib/goal-waterfall.ts

computeGoalWaterfall(goals, savedPrincipalCents, committedPrincipalCents) — no DB calls, no side effects. Goals are processed in sort_order ascending. For each goal:

- savedAllocatedCents = min(goal.targetPrincipalCents, remainingSaved)
- committedAllocatedCents = min(goal.targetPrincipalCents, remainingCommitted)
- savedPercent = savedAllocated / target × 100
- fundingState = 'funded' | 'partially_funded' | 'unfunded' (based on savedAllocated vs target)

surplusCents = max(0, savedPrincipalCents " sum of all active goal targets). unallocatedCents is computed at the route level as max(0, jar.goalAmountCents " activeGoalTargetSum).

6. Key Invariants Enforced

- Goal mutations create ZERO ledger_entries and ZERO financial_transactions (verified by PART I tests)
- savedPrincipalCents = refundablePrincipalCents + committedPrincipalCents (fees and refunds excluded)
- committedPrincipalCents "d savedPrincipalCents at all times (invariantHolds flag in financial summary)
- SUM(active goal targets) "d jars.goalAmountCents — enforced inside SELECT...FOR UPDATE transaction
- Archived goals excluded from waterfall and from the active target sum used in constraint checks
- Hard delete only in Draft/Inviting; soft archive (preserves history) once jar has launched
- All allocation is derived at query time — targetPrincipalCents is the only stored goal amount
- Members cannot see other members' exact dollar balances — only their own + statusLabel for others

7. Mobile UI (Phase 4D)

New Hooks

artifacts/mobile/hooks/useJarGoals.ts

useJarGoals(jarId, options) — queries GET /jars/:jarId/goals via customFetch (auto-authenticated). Enabled only when the Overview tab is active. Returns goals[], unallocatedCents, surplusCents, savedPrincipalCents, committedPrincipalCents, goalAmountCents.

artifacts/mobile/hooks/useFinancialSummary.ts

useFinancialSummary(jarId, options) — queries GET /jars/:jarId/financial-summary. Returns full JarFinancials, GoalAllocation[], MemberFinancialSummary[], viewerRole, invariantHolds.

Jar Detail Screen — artifacts/mobile/app/jar/[id].tsx

renderOverview() updated to render two new sections below the action row:

- Goals Section: shows each active goal with a ProgressBar, funding-state icon (Feather), saved/target amounts, and a 'Manage' link for organizers. Unallocated bucket shown when > 0.
 - Financial Details accordion: collapsed by default, expands to show refundable/locked/pending/refunded/fee breakdown. Uses Feather chevron-up/down icon. Over-target warning shown in amber if applicable.
- onRefresh() updated to concurrently refetch goals and financial summary on Overview tab pull-to-refresh.

8. Test Coverage

phase4d-goals.test.ts ("H 42 tests across 10 PARTS)

- A GET goals — empty state: structure, unallocated = jar target, surplus = 0
- B POST goals — create, auto-increment sortOrder, sum constraint (409), field validation
- C Authorization — member read-only, outsider 403, unauthenticated 401, organizer-only mutations
- D PATCH goal — name/desc/target updates, sum constraint, 404 for unknown goal
- E Archive — hard delete (Draft jar), soft archive (Saving jar), 404 on re-archive of deleted
- F Reorder — canonical sort_order 1...N, reject incomplete list, duplicates, invalid IDs, empty array
- G Waterfall algebra — all unfunded with zero principal, unallocated math, surplus = 0, ascending order
- H Archived goals excluded — active list shrinks, unallocatedCents recalculates, includeArchived param
- I Invariant — POST/PATCH/reorder/archive create zero ledger_entries and zero financial_transactions
- J PATCH jar amount constraint — allow increase, reject < goal sum (409), allow exactly = goal sum

phase4d-transparency.test.ts ("H 30 tests across 6 PARTS)

- A Zero state — top-level shape, all totals = 0, remainingToSave = goalAmountCents, goals = []
- B Authorization — organizer 200, active member 200, outsider 403, unauthenticated 401
- C Privacy model — organizer sees exact figures for all; member sees own exact + others redacted
- D Goal waterfall — all goals unfunded at zero principal, unallocatedCents math, endpoint parity
- E Jar-level totals — remainingToSaveCents, isOverTarget = false at zero state, currency propagation
- F Ledger-backed totalSaved — GET /jars/:jarId and GET /jars return totalSavedCents=0 with no real payments

Full Suite Result

19 test files | 501 tests | 501 passed | 0 failed

9. Delivery Checklist

- ' Migration 0018 (jar_goals) applied to live database
- ' meta/_journal.json updated with idx:18 entry
- ' lib/db rebuilt; jarGoals exported from @workspace/db
- ' jarGoals table definition + jarGoalsRelations in schema/index.ts
- ' computeGoalWaterfall() pure function — no DB calls, no side effects
- ' getJarSavedPrincipalCents() replaces getTotalSaved() — all 3 call-sites updated
- ' goals.ts — all 6 endpoints with correct auth, validation, constraint enforcement
- ' financial-summary.ts — privacy-scoped member breakdowns + goal waterfall
- ' jars.ts PATCH — goal sum constraint check added
- ' routes/index.ts — goalsRouter and financialSummaryRouter registered
- ' ActivityEventType extended: goal_created, goal_updated, goal_archived, goal_reordered
- ' useJarGoals.ts + useFinancialSummary.ts mobile hooks created

- ' jar/[id].tsx — Goals section + Financial Details accordion in renderOverview
- ' onRefresh() includes concurrent refetch for goals and financial summary
- ' phase4d-goals.test.ts — 10 test parts, all passing
- ' phase4d-transparency.test.ts — 6 test parts, all passing
- ' api-server typecheck: CLEAN
- ' mobile typecheck: CLEAN
- ' lib/db typecheck: CLEAN
- ' Full test suite: 501/501 passing
- ' Committed to main: 141523e
- ' Pushed to origin/main (JB102298/TripJar)

10. Files Changed

- [NEW] lib/db/drizzle/0018_phase4d_jar_goals.sql
- [NEW] artifacts/api-server/src/lib/goal-waterfall.ts
- [NEW] artifacts/api-server/src/routes/goals.ts
- [NEW] artifacts/api-server/src/routes/financial-summary.ts
- [NEW] artifacts/api-server/src/__tests__/phase4d-goals.test.ts
- [NEW] artifacts/api-server/src/__tests__/phase4d-transparency.test.ts
- [NEW] artifacts/mobile/hooks/useJarGoals.ts
- [NEW] artifacts/mobile/hooks/useFinancialSummary.ts
- [MOD] lib/db/drizzle/meta/_journal.json
- [MOD] lib/db/src/schema/index.ts
- [MOD] artifacts/api-server/src/lib/activity.ts
- [MOD] artifacts/api-server/src/lib/financial-balance.ts
- [MOD] artifacts/api-server/src/routes/jars.ts
- [MOD] artifacts/api-server/src/routes/index.ts
- [MOD] artifacts/mobile/app/jar/[id].tsx